# Di Feng

MD/PhD in Molecular Immunology and Experimental Pathology

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## Experience

**Senior Principal Scientist — Computational Innovation, Oncology** Sept. 2024 – Present
*Boehringer Ingelheim Pharmaceuticals* *Ridgefield, CT*

- Led development of an AI agent platform for oncology target discovery and indication prioritization, serving as a Forward Deployed Engineer (FDE) bridging TA research and Computational innovation.
- Built evaluation frameworks (Evals) for LLM, LangGraph-based agents, enabling robust multi-step reasoning and benchmarking performance in scientific hypothesis generation tasks.
- Served as portfolio project lead, developing computational data packages integrating multi-omics and real-world oncology data (RWD) to predict and prioritize novel therapeutic targets.
- Drove cross-department collaboration to co-create novel drug targets, leveraging TME immune archetype analysis and intercellular communication networks.
- Mentored and collaborated with Yale postdoctoral researchers to apply CRISPR and machine learning for identifying novel genes regulating cancer immunity.

**Senior Principal Scientist — Computational Immunology** Oct. 2019 – Sept. 2024
*Boehringer Ingelheim Pharmaceuticals* *Ridgefield, CT*

- Designed regulatory safety pipelines for viral modeling, molecular simulations, and genomic risk analysis.
- Led data analysis for research collaboration with Icahn School of Medicne at Mount Sinai for cancer therapies.
- Managed AI/ML partnership with Case Western Center for Computational Imaging and Personal Diagnostics.
- Led the development of histopathology image analytics to evaluate drug efficacy in preclinical models.
- Developed spatial image biomarkers based on immune cell spatial patterns associated with prognosis of cancer.
- Created drug target evaluation framework to improve antibody, small molecule, and viral portfolio advancement.

**Principal Scientist — Computational Expert Lead** Mar. 2018 – July 2019
*Boehringer Ingelheim Pharmaceuticals* *Ridgefield, CT*

- Led the development of a single cell analysis software facilitating collaboration and sharing of results between computational biologists and experimental scientists.
- Built web data portal for analyzing and visualizing patients' RNA-Seq and Microassay gene expression datasets
- Managed a team of computational scientists on bioinformatics projects spanning oncology, immunology, fibrosis, and cardio-metabolism drug discovery

**Principal Scientist — Integrative Biology** Nov. 2013 – Mar. 2018
*Boehringer Ingelheim Pharmaceuticals* *Ridgefield, CT*

- Bioinformatics liaison for Immunology Department - worked with dept. head to align & execute scientific strategies
- Built NGS capabilities with software development teams. Developed and deployed data infrastructure, databases, and software for data ingestion, processing, and bioinformatics analysis.
- Identified genomic biomarkers and disease indication finding for including Spesolimab and Skyrizi.
- Guided sequence based experiential design for multiple drug development programs.

**Postdoctoral Researcher** Feb 2006 – Oct 2013
*Cancer Institute of New Jersey* *Newark, NJ*

- Led the research using molecular biochemistry methods and mouse models to dissect correlation of IRF5 gene with lupus susceptibility, integrating bioinformatics with wet lab science.
- Designed cell based assay to screen anti-inflammatory compounds in collaboration with Takeda Pharmaceuticals.
- Created patented imaging algorithms for nuclear autoantibody staining pattern analysis in lupus diagnostics.
- Investigated the role of IRF5 in human B cells from lupus patients, supported by the Lupus Research Alliance.
- Examined the short & long term effects of space radiation exposure on animals' immunological response systems.

**Graduate Research Assistant** Sept 2000 – Jan 2006
*Rutgers University* *Newark, NJ*

- Conducted NIH-sponsored(R01) AIDS clinical research, studying innate immunity dysfunction in HIV infection.
- Developed bioassay for testing anti-terrorism air decontamination system with Plasmasol Corporation.
- Performed fundamental research of dendritic cell responses to viral pathogens (HIV, HBV, HCV, HSV)

## Technical Skills

**Languages**: Bash/Unix shell, Python,R/Bioconductor/Rshiny, JavaScript
**Technologies**: PyTorch, LLM, LangChain, LangGraph, OpenCV, CNN, Transformers, AWS, Databricks, HPC, Git
**Bioinformatics**:Multi-Omics,Spatial, Seurat, Scanpy, ScVI,GSEA,foundation model, DeepChem,RDKit,
**Wet Lab:** Next Generation Sequencing, cell culture, PBMC isolation, cell purification (MACS), Flow Cytometry, Imaging flow, ELISA, Western blot, IHC, PCR, molecular cloning, gene silencing, co-immunoprecipitation, virus culture, viral staining, plaque assay, interferon bioassay
**Concepts**: Lab-in-the-loop,Machine Learning, Computer Vision,Cloud Computing, Causality, Collective intelligence

## Education

**Rutgers University** — Newark, NJ
*Ph.D. Molecular Immunology, Experimental Pathology* — *2000 – 2006*

**School of Medicine, Jiaotong University,** — Shanghai, CN
*B.A., M.S. of Medicine (equivalent to MD)* — *1991 – 1998*

## Awards

- Alliance of Lupus Pilot Grant: Identifying IRF5-Mediated Pathways In Normal And SLE B Cells, 2012
- 2008 Dolph Adams Award for the Most Highly Cited Paper in Five Years in the Journal of Leukocyte Biology https://doi.org/10.1189/jlb.0908999

## Recent External talks

- Yale University-YSPH Biostatistics Seminar:Unlocking the Full Potential of Image Data in Research: The Journey from Analytics to Artificial Intelligence, 28 March 2023

## Technical Skills

**Programming and tools**:Bash/Unix shell, R/Bioconductor/Rshiny, Python, multiple ML libraries, cloud and high-performance computing environments.
**Image analysis**:Computer vision, Deep learning model of histopathology and medical image data for object detection, segmentation, feature extraction, and integrating multi-modal genomics data and clinical information. Proficient in designing and optimizing convolutional neural networks (CNNs) and other deep learning models for image segmentation, classification and regression tasks.Assessing ML/AI model performance using metrics. Spatial relationships model. Handling large-scale image datasets, including organization, storage, and retrieval.Visualization: Competent in visualizing image analysis results to effectively communicate findings. Python libraries: OpenCV, Pillow, scikit-image, PyTorch, Mahotas, Weights & Biases MLOps Platform
**NLP**:In-context learning, OpenLLaMA, Lit-LLaMA,Colossal-AI, RedPajama-Data, ALPACA
**Bioinformatics**: Multi-Omics NGS data analysis including bulk, spatial (Visium, GeoMx, CosMx), single cell genomics or multi-omics (10XGenomic cell ranger, space ranger, Seurat, Scanpy, Monocle, Harmony, OmicVerse,scverse, ScVI, scANVI, totalVI), NGS analysis: FASTQC, Bowtie, BWA, SAM tools, BED tools, STAR, variant callers, SNPs, indels identification GATK HaplotypeCaller, Mutect2, VarScan, Strelka. Transposon: TIPseq, RNA-seq, ATAC-seq, CHIP-seq, CITE-seq, variant/somatics, variants annotation including ANNOVAR, SnpEff, Variant Effect Predictor (VEP).Flow cytometry or CyTOF using viSNE, SPADE, Citrus and FlowSOM.Proteomics, and metabolites data analysis, KNIME Platform,NextFlow.
**Genomics**:GWAS, TWAS, Mendelian randomization analysis, MAGMA.
**Protein modeling and structural biology** :Protein structure prediction and molecule docking, and generative model for protein design, molecular simulation:ESMFold, DiffDock, ProtGPT2, openMM, ACEMD
**Chemoinformatics**: Molecule and drug design, chemical structure representation and searching. Open Babel,ChemDraw,DeepChem,RDKit,OpenEye Toolkits, MegaMolBART
**Statistical analysis**: Analysis of multivariate and high-dimensional data, dimensional reduction, clustering, feature and variable selection selection (lasso, elastic net, and random forest, multivariable analysis, meta analysis, fixed-effects ,random-effects models, mixed effects models, multiple regression model, imputation, cross validation, time to event analysis (survival analysis), longitudinal data analysis, ML/AI classification and regression model, supervised/unsupervised learning, ensemble models,bagging and boosting, causal analysis .
**Molecular analysis**: Clone and protein expression codon optimization. NGS experiment design and data generation from cell lines, primary cells, organoids, patient explants, tissue slice, patient-derived cancer models, histology, and image analysis for biomarker discovery.

**Data**:GEO, TCGA, CPTAC, IMGT, UCSC Xena, Ensembl, CMap/CLUE, cBioportal, HPA, Blueprint, GTEX, CCLE, gnomAD Genome Aggregation Database, Cancer Dependency Map (DepMap), dbSNP, 1000 Genomes, COSMIC, HGMD ClinVar, Matrisome, surfaceome, GO (Gene Ontology), Hallmark, Reactome, WikiPathways, MSigDB gene set, EU Innovative Medicines Initiative (IMI), Biobank FinnGen, UK biobank, Rare Disease: Orphanet, NORD, GARD, Rare Disease Database, EURODIS.
**Software**: Omicsoft, Genedata.

## PUBLICATIONS


[1] Kaiyang Tang et al. "Perturbation of calreticulin potentiates CD8+ T cell-mediated antitumor immunity". In: *Journal of Experimental Medicine* (2025). DOI: https://doi.org/10.1084/jem.20242360.

[2] Chuanpeng Dong et al. "TCPGdb: A comprehensive CRISPR screen database of T cell perturbation genomics for identification of critical T cell regulators". In: *Cancer Immunology Research* (2025). DOI: https://doi.org/10.1158/2326-6066.CIR-25-0168.

[3] Chuanpeng Dong et al. "Sensitive detection of synthetic response to cancer immunotherapy driven by gene paralog pairs". In: *Patterns* (2025). DOI: https://doi.org/10.1016/j.patter.2025.101184.

[4] Amit Weiss et al. "A Deep Learning Framework for Causal Inference in Clinical Trial Design: The Clinical Trials Uncovering Real Efficacy Artificial Intelligence Large Clinicogenomic Foundation Model". In: *AI in Precision Oncology* (2025). DOI: https://doi.org/10.1177/2993091X251369852.

[5] Rong Li et al. "Incorporating prior information in gene expression network-based cancer heterogeneity analysis". In: *Biostatistics* 26.1 (2025), kxae028. DOI: https://doi.org/10.1093/biostatistics/kxae028.

[6] Robert J Norgard et al. "Reshaping the tumor microenvironment of KRASG12D pancreatic ductal adenocarcinoma with combined SOS1 and MEK inhibition for improved immunotherapy response". In: *Cancer Research Communications* 4.6 (2024), pp. 1548–1560. DOI: https://doi.org/10.1158/2767-9764.crc-24-0172.

[7] Dechao Shan et al. "Deep learning for discovering pathological continuum of crypts and evaluating therapeutic effects: An implication for in vivo preclinical study". In: *Plos one* 16.6 (2021), e0252429. DOI: https://doi.org/10.1371/journal.pone.0252429.

[8] Di Feng et al. "Single Cell Explorer, collaboration-driven tools to leverage large-scale single cell RNA-seq data". In: *BMC genomics* 20 (2019), pp. 1–8. DOI: https://doi.org/10.1186/s12864-019-6053-y.

[9] Samantha A Chalmers et al. "Remission induction of established nephritis by therapeutic administration of a novel BTK inhibitor in an inducible model of lupus nephritis". In: *The Journal of Immunology* 198.1_Supplement (2017), pp. 224–16. DOI: https://doi.org/10.4049/jimmunol.198.supp.224.16.

[10] Sammy Chalmers et al. *II-06 Therapeutic blockade of immune complex-mediated glomerulonephritis by highly selective inhibition of bruton's tyrosine kinase.* 2016. DOI: https://doi.org/10.1136/lupus-2016-000179.36.

[11] SA Chalmers et al. "OP0164 Blockade of Immune Complex-Mediated Glomerulonephritis by Highly Selective Inhibition of Bruton's Tyrosine Kinase". In: *Annals of the Rheumatic Diseases* 75 (2016), pp. 117–118. DOI: https://doi.org/10.1136/annrheumdis-2016-eular.3649.

[12] Samantha A Chalmers et al. "Therapeutic blockade of immune complex-mediated glomerulonephritis by highly selective inhibition of Bruton's tyrosine kinase". In: *Scientific reports* 6.1 (2016), p. 26164. DOI: https://doi.org/10.1038/srep26164.

[13] Erica Maria Pimenta et al. "IRF5 is a novel regulator of CXCL13 expression in breast cancer that regulates CXCR5+ B-and T-cell trafficking to tumor-conditioned media". In: *Immunology and cell biology* 93.5 (2015), pp. 486–499. DOI: https://doi.org/10.4049/jimmunol.192.supp.203.37.

[14] X Bi et al. "Deletion of Irf5 protects hematopoietic stem cells from DNA damage-induced apoptosis and suppresses $\gamma$-irradiation-induced thymic lymphomagenesis". In: *Oncogene* 33.25 (2014), pp. 3288–3297. DOI: https://doi.org/10.1038/onc.2013.295.

[15] Saurav De et al. “Targeting IRF5 inhibition in human B cells: identification of new functional roles that implicate IRF5 in systemic lupus erythematosus B-cell pathology”. In: *Arthritis Research & Therapy* 16 (2014), pp. 1–23. DOI: https://doi.org/10.1186/ar4632.

[16] Erica Pimenta et al. “IRF5 is a novel regulator of CXCL13 expression in breast cancer that increases CXCR5+ B and T cell trafficking to the tumor (TUM7P. 955)”. In: *The Journal of Immunology* 192.1_Supplement (2014), pp. 203–37. DOI: https://doi.org/10.4049/jimmunol.192.supp.203.37.

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[18] Di Feng and Betsy J Barnes. “Bioinformatics analysis of the factors controlling type I IFN gene expression in autoimmune disease and virus-induced immunity”. In: *Frontiers in Immunology* 4 (2013), p. 291. DOI: https://doi.org/10.3389/fimmu.2013.00291.

[19] Rivka C Stone et al. “RNA-Seq for enrichment and analysis of IRF5 transcript expression in SLE”. In: *PloS one* 8.1 (2013), e54487. DOI: https://doi.org/10.1371/journal.pone.0054487.

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[21] Di Feng et al. “Protection of Irf5-deficient mice from pristane-induced lupus involves altered cytokine production and class switching”. In: *European journal of immunology* 42.6 (2012), p. 1477. DOI: https://doi.org/10.4049/jimmunol.188.supp.159.11.

[22] Di Feng et al. “I rf5-deficient mice are protected from pristane-induced lupus via increased T h2 cytokines and altered I g G class switching”. In: *European Journal of Immunology* 42.6 (2012), pp. 1477–1487. DOI: https://doi.org/10.1002/eji.201141642.

[23] Rivka C Stone et al. “Interferon regulatory factor 5 activation in monocytes of systemic lupus erythematosus patients is triggered by circulating autoantigens independent of type I interferons”. In: *Arthritis & Rheumatism* 64.3 (2012), pp. 788–798. DOI: https://doi.org/10.1002/art.33395.

[24] Rivka C Stone et al. “IRF5 activation in monocytes of SLE patients is triggered by circulating autoantigens independent of type I IFN”. In: *Arthritis and Rheumatism* 64.3 (2012), p. 788. DOI: https://doi.org/10.1016/j.cyto.2011.07.400.

[25] Lisong Yang et al. “Monocytes from Irf5-/- mice have an intrinsic defect in their response to pristane-induced lupus”. In: *The Journal of Immunology* 189.7 (2012), pp. 3741–3750. DOI: https://doi.org/10.4049/jimmunol.1201162.

[26] Rivka Stone et al. “Determination of the Contribution of An IRF5-SLE Risk Haplotype to IRF5 expression and Alternative Splicing Using Next-Generation Sequencing”. In: *ARTHRITIS AND RHEUMATISM*. Vol. 63. 10. WILEY-BLACKWELL COMMERCE PLACE, 350 MAIN ST, MALDEN 02148, MA USA. 2011, S966–S967. DOI: https://doi.org/10.3410/f.1025508.374958.

[27] Rivka Stone et al. “IRF5 activation in monocytes of SLE patients is triggered by circulating autoantigens independent of type I IFN”. In: *ARTHRITIS AND RHEUMATISM*. Vol. 63. 10. WILEY 111 RIVER ST, HOBOKEN 07030-5774, NJ USA. 2011, S559–S559. DOI: https://doi.org/10.1016/j.cyto.2011.07.400.

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[32] Rivka C Stone et al. "PS1-52 Interferon regulatory factor 5 (IRF5) activation in systemic lupus erythmatosus". In: *Cytokine (Philadelphia, Pa.)* 52.1-2 (2010), pp. 28–28. DOI: https://doi.org/10.1016/j.cyto.2010.07.124.

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[38] Stacey L Fanning et al. "Receptor cross-linking on human plasmacytoid dendritic cells leads to the regulation of IFN-$\alpha$ production". In: *The Journal of Immunology* 177.9 (2006), pp. 5829–5839. DOI: https://doi.org/10.4049/jimmunol.177.9.5829.

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[40] JH Dai et al. "The role of IRF-7 in IFN-alpha production by human plasmacytoid dendritic cells". In: *Faseb Journal*. Vol. 17. 7. FEDERATION AMER SOC EXP BIOL 9650 ROCKVILLE PIKE, BETHESDA, MD 20814-3998 USA. 2003, pp. C295–C295. DOI: https://doi.org/10.4049/jimmunol.186.supp.111.30.

[41] Alexander Izaguirre et al. "Comparative analysis of IRF and IFN-alpha expression in human plasmacytoid and monocyte-derived dendritic celis". In: *J Leukoc. Bio)* 74 (2003), pp. 1–13. DOI: https://doi.org/10.1189/jlb.0603255.